BUSI 246 - Data Analysis Project

S&P 500 and its 11 underlying industries within

Intro

For our project we looked at the closing price of the S&P 500 and compared it to the 11 different industries within the index. We took a sample of data starting on January 5, 2024 and ending on October 3, 2025.
We looked at descriptive statistics, correlation, and ANOVA.
Finally, we made a hypothetical chart showing what would have happen if someone invested $100 into each of the industries starting January 5, 2024.

Variables and research questions

Variables:

- Output -
  - Descriptive Statistics
  - Anova Single Factor
  - Backtesting
- Input - Data from the S&P 500 and the underlying indexes/industries

Research Question: How do the performance trends of individual industries within the S&P 500 correlate with the overall performance of the index, and which industries have the most significant impact on the index's movements? Finally, which industry we recommend given your risk profile.

S&P500		HealthCare		Technology		Energy	
Mean	5694.35	Mean	1664.18	Mean	4368.22	Mean	676.68
Standard Error	50.11	Standard Error	7.94	Standard Error	55.89	Standard Error	3.32
Median	5694.61	Median	1678.31	Median	4379.66	Median	676.59
Mode	#N/A	Mode	#N/A	Mode	#N/A	Mode	#N/A
Standard Deviation	480.63	Standard Deviation	76.12	Standard Deviation	536.08	Standard Deviation	31.86
Sample Variance	231006.35	Sample Variance	5794.83	Sample Variance	287376.46	Sample Variance	1015.15
Kurtosis	-0.69	Kurtosis	-0.69	Kurtosis	-0.35	Kurtosis	-0.47
Skewness	0.08	Skewness	0.05	Skewness	0.31	Skewness	-0.10
Range	2018.55	Range	314.39	Range	2401.85	Range	143.46
Minimum	4697.24	Minimum	1515.32	Minimum	3259.48	Minimum	605.93
Maximum	6715.79	Maximum	1829.71	Maximum	5661.33	Maximum	749.39
Sum	523880.08	Sum	153104.20	Sum	401876.48	Sum	62254.85
Count	92.00	Count	92.00	Count	92.00	Count	92.00

Finance		Consumer Discretionary		Communication Services		Industrials	
Mean	774.20	Mean	1626.51	Mean	326.88	Mean	1125.14
Standard Error	8.54	Standard Error	18.34	Standard Error	4.51	Standard Error	10.01
Median	781.80	Median	1586.10	Median	319.51	Median	1123.99
Mode	#N/A	Mode	#N/A	Mode	318.99	Mode	#N/A
Standard Deviation	81.91	Standard Deviation	175.88	Standard Deviation	43.24	Standard Deviation	96.05
Sample Variance	6709.89	Sample Variance	30932.37	Sample Variance	1869.30	Sample Variance	9224.82
Kurtosis	-1.36	Kurtosis	-1.36	Kurtosis	-0.22	Kurtosis	-0.86
Skewness	-0.15	Skewness	0.29	Skewness	0.54	Skewness	0.22
Range	274.98	Range	572.54	Range	195.39	Range	367.04
Minimum	625.08	Minimum	1369.16	Minimum	243.15	Minimum	944.00
Maximum	900.06	Maximum	1941.70	Maximum	438.54	Maximum	1311.04
Sum	71225.98	Sum	149638.79	Sum	30072.74	Sum	103512.96
Count	92.00	Count	92.00	Count	92.00	Count	92.00

Consumer Staples		Utilities		Real Estate		Materials	
Mean	856.92	Mean	383.19	Mean	256.92	Mean	560.87
Standard Error	4.39	Standard Error	3.93	Standard Error	1.38	Standard Error	2.51
Median	868.73	Median	396.23	Median	259.98	Median	563.19
Mode	#N/A	Mode	#N/A	Mode	#N/A	Mode	#N/A
Standard Deviation	42.07	Standard Deviation	37.68	Standard Deviation	13.28	Standard Deviation	24.08
Sample Variance	1770.12	Sample Variance	1419.99	Sample Variance	176.36	Sample Variance	580.07
Kurtosis	-0.68	Kurtosis	-0.72	Kurtosis	-0.64	Kurtosis	-0.01
Skewness	-0.68	Skewness	-0.56	Skewness	-0.24	Skewness	-0.32
Range	155.79	Range	145.77	Range	57.67	Range	124.95
Minimum	762.55	Minimum	305.72	Minimum	224.97	Minimum	492.02
Maximum	918.34	Maximum	451.49	Maximum	282.64	Maximum	616.97
Sum	78836.97	Sum	35253.51	Sum	23636.90	Sum	51600.43
Count	92.00	Count	92.00	Count	92.00	Count	92.00

Descriptive Statistics

Based on the descriptive statistics, the S&P 500 and Technology sectors have the highest average values, suggesting stronger overall performance for this sector.

In addition looking at the Standard Deviation we can see how volatile each of these indexes are.

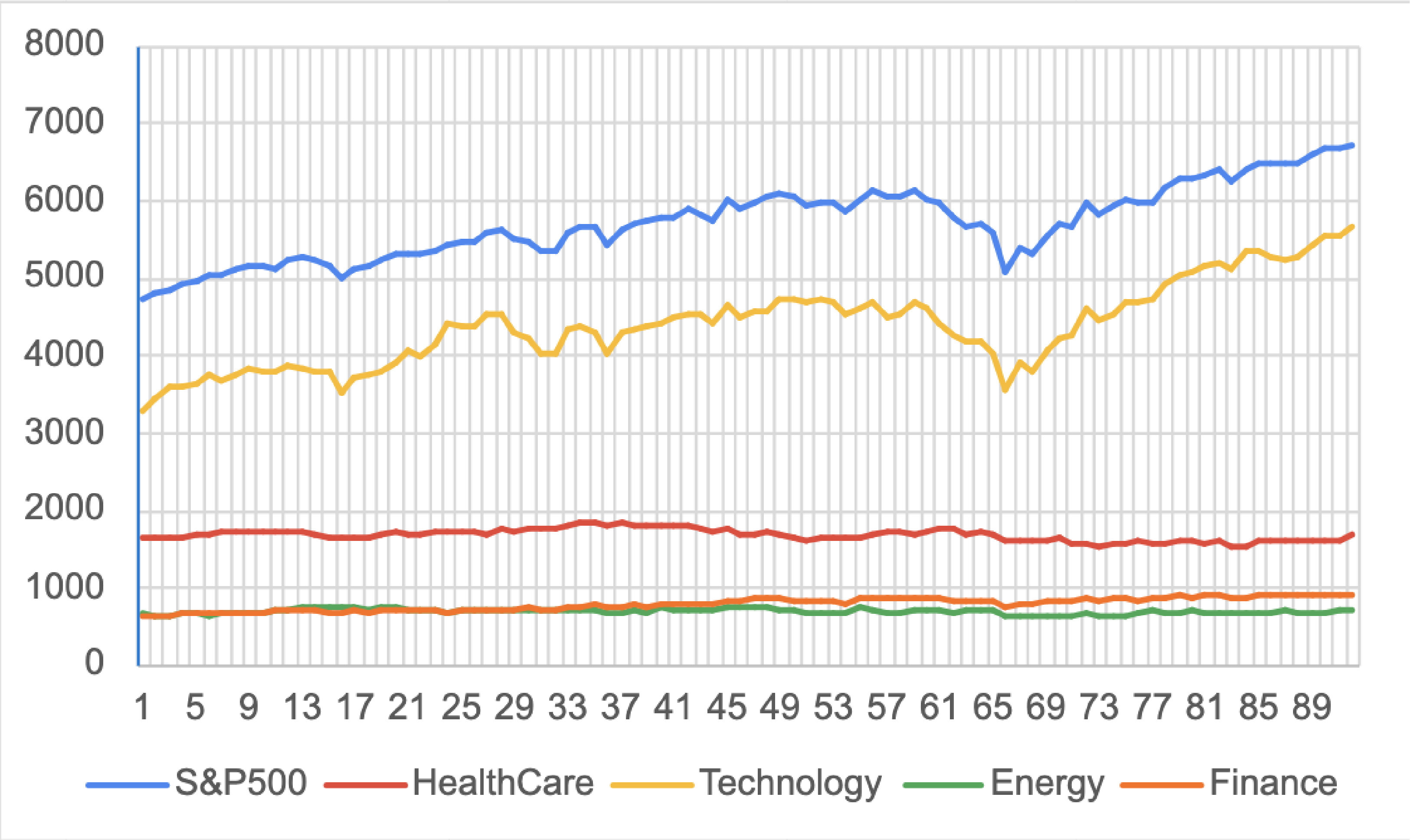
High -

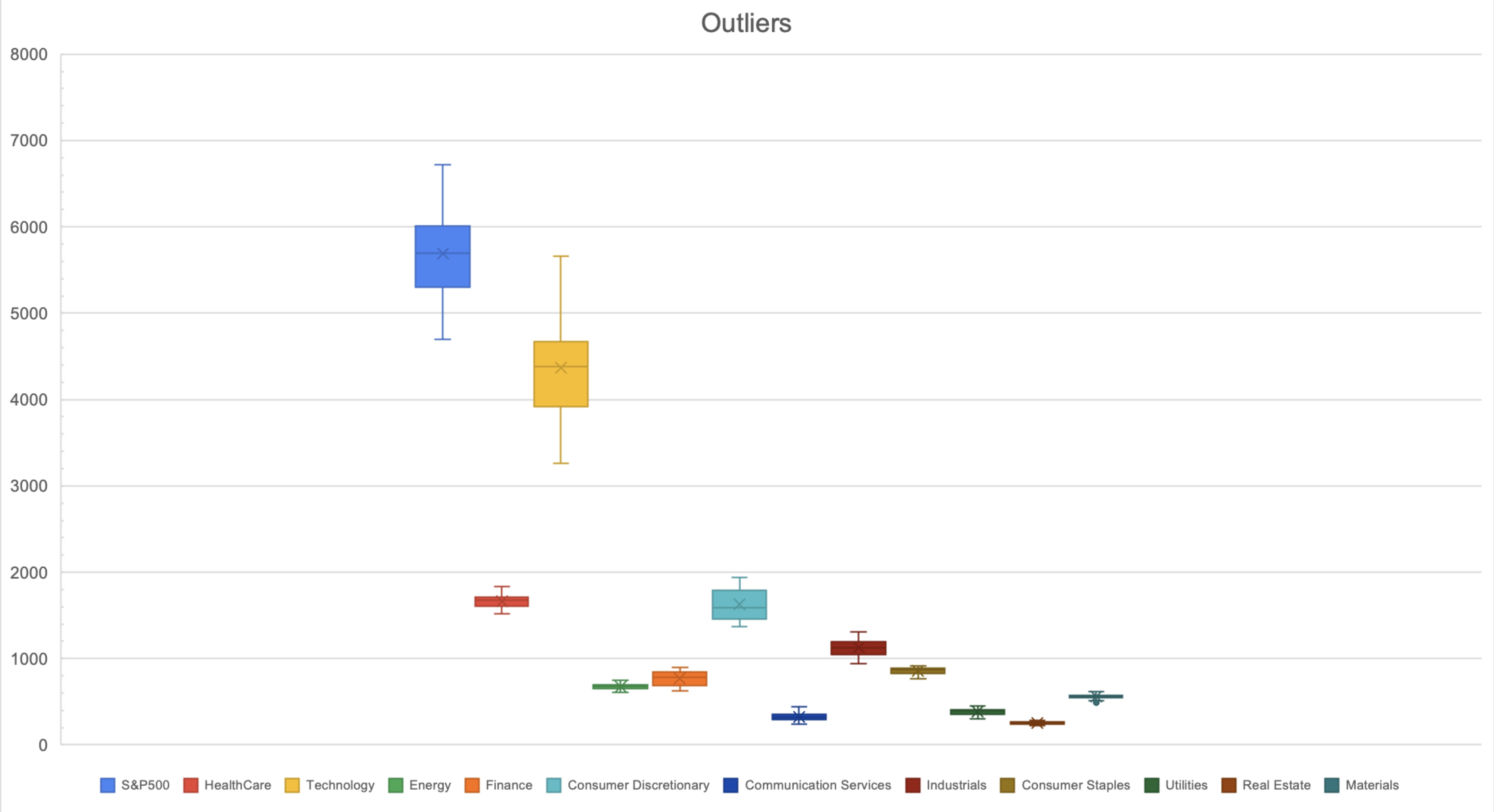
- Technology - 536
- Consumer Discretionary- 175

Lowest -

- Real Estate - 13
- Materials - 24

Price Fluctuations Over Time





Outliers
8000
7000
6000
5000
4000
3000
2000
1000
0
S&P500
HealthCare
Technology
Energy
Finance
Consumer Discretionary
Communication Services
Industrials
Consumer Staples
Utilities
Real Estate
Materials

Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
S&P500	92	523880.08	5694.348696	231006.3523
HealthCare	92	153104.2	1664.176087	5794.833967
Technology	92	401876.48	4368.222609	287376.4551
Energy	92	62254.85	676.6831522	1015.150457
Finance	92	71225.98	774.1954348	6709.888344
Consumer Discretionary	92	149638.79	1626.508587	30932.36825
Communication Services	92	30072.74	326.8776087	1869.303651
Industrials	92	103512.96	1125.14087	9224.822892
Consumer Staples	92	78836.97	856.923587	1770.121184
Utilities	92	35253.51	383.1903261	1419.985625
Real Estate	92	23636.9	256.9228261	176.3635436
Materials	92	51600.43	560.8742391	580.0666005

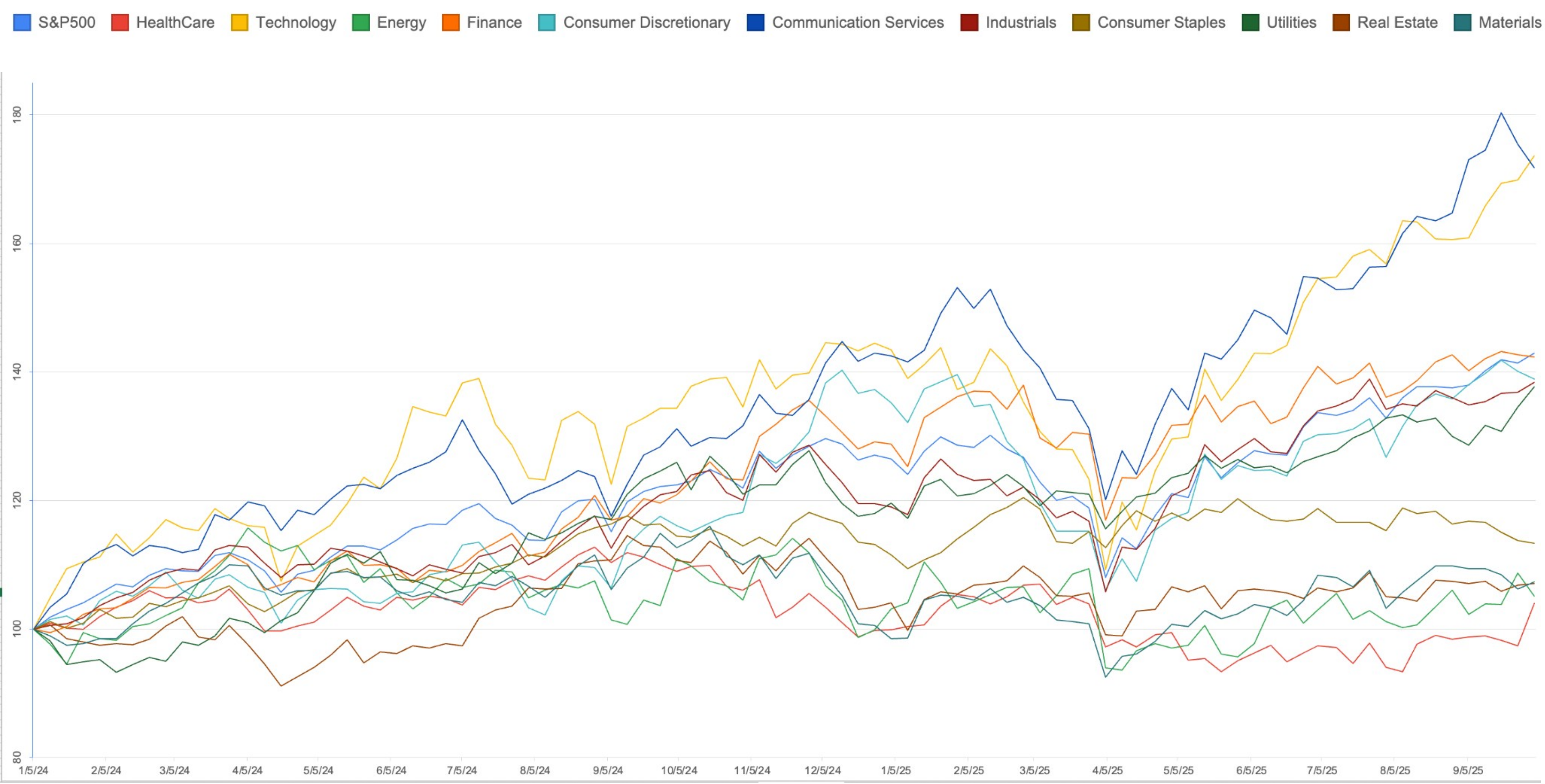
ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	3005031698	11	273184699.9	5672.874514	0.0000	1.797404342
Within Groups	52586689.79	1092	48156.30933			
Total	3057618388	1103				

Anova Single Factor Hypothesis

We reject the null hypothesis because the p-value is zero.

We say that there is difference between industries.



S&P500
HealthCare
Technology
Energy
Finance
Consumer Discretionary
Communication Services
Industrials
Consumer Staples
Utilities
Real Estate
Materials
180
160
140
120
100
80
1/5/24
2/5/24
3/5/24
4/5/24
5/5/24
6/5/24
7/5/24
8/5/24
9/5/24
10/5/24
11/5/24
12/5/24
1/5/25
2/5/25
3/5/25
4/5/25
5/5/25
6/5/25
7/5/25
8/5/25
9/5/25

Industry Performance

The previous chart shows what happens when someone put $100 into each sector on January 5, 2024.

The highest performance would be for communication services and technology with technology giving a 73% return and communication services giving 71% return.

Then medium performance would be finance, industrials, utilities, as well as the whole S&P 500 overall.

Finally, the lowest performing sectors are consumer staples, materials, energy, healthcare, and real estate.

Conclusion

Research Question: How do the performance trends of individual industries within the S&P 500 correlate with the overall performance of the index, and which industries have the most significant impact on the index's movements? Finally, which industry we recommend given your risk profile.

As we can on the chart of price fluctuations, the technology industry is the one that is the most correlated to the S&P 500 and the rest industries don't fluctuate as much over time, therefore contributing less to the overall performance of the S&P 500. We can agree that the S&P 500 growth is highly dependent on the performance of technology companies.

Risk profile:

- Low: low volatility industries such as consumer staples (outperformed other peers: materials, energy, healthcare, and real estate)
- Medium: medium volatility industries - finance (outperforming its other peers: industrials, utilities, S&P 500)
- High: high risk tolerance - technology
    - However tech performance is very close to communication services, and communication services has less volatility